

Ch 9: Light Reflection & Refraction.

One Word Answer:-

Q1) You are given kerosene, turpentine and water. In which of these does the light travel fastest?

Use the information given in table 9.3.

Ans:- Light travel fastest in water, because among the three, water has the least refractive index. Higher refractive index means slower speed of light in that medium, so the medium with the lowest refractive index lets light travel fastest.

Q2) No matter how far you stand from a mirror, your image appears erect. The mirror is likely to be -

- a) Only plane.
- b) Only concave.
- c) Only convex
- d) Either plane or convex.

Ans:- Both plane or convex.

Both plane mirrors and convex mirrors always form a virtual, erect image regardless of the object's distance.

Q.31 One - half of a convex lens is covered with a black paper. Will this lens produce a complete image of the object?

Verify your answer experimentally. Explain your observations.

Ans. Yes, the lens will still produce a complete image of the object. This is because every point of the object sends light rays in all directions, and even the uncovered half of the lens receives enough rays from every point to form a full image - however, the intensity (brightness) of the image will be reduced, since less light now passes through the lens.

Short Answer:

Q.41 We wish to obtain an erect image of an object, using a concave mirror of focal length 15 cm. What should be the range of distance of the object from the mirror? What is the nature of the image? Is the image larger or smaller than the object? Draw a ray diagram to show the image formation in this case.

Ans) Range of the distance of the object from the mirror is $0 < u < 15$ cm, i.e. u will be between 0 cm and 15 cm. The nature of the image = virtual. Size of the image = larger than the object.

Q5) An object 5 cm in length is held 25 cm away from a converging lens of focal length 10 cm. Draw the ray diagram and find the position, size and the nature of image formed.

Ans) Given $h = 5$ cm, $u = -25$ cm, $f = +10$ cm
Using the lens formula:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} + \frac{1}{-25} = \frac{5-2}{50} = \frac{3}{50}$$

$$v = \frac{50}{3} = 16.7 \text{ cm (approx)}$$

Magnification:

$$m = \frac{v}{u} = \frac{50/3}{-25} = -2$$

$$h' = m \times h = \frac{-2}{3} \times 5 = -3.3 \text{ cm}$$

Results:

- Position of image : 16.7 cm from the lens.
- Size of image : 3.3 cm.
- Nature of image : Real and inverted.

Long Answer:

Q6] Light enters from air to glass having refractive index 1.50. What is the speed of light in glass? Speed of light in vacuum is $3 \times 10^8 \text{ ms}^{-1}$.

Ans) Given that speed of light in vacuum i.e., the speed of light in air ($c = 3 \times 10^8 \text{ ms}^{-1}$) and the refractive index of glass $n_g = 1.5$. Since refractive index $n_g = \frac{\text{Speed of light in air}}{\text{Speed of light in glass}}$.

$$\Rightarrow n_g = \frac{c}{v_g} \Rightarrow 1.5 = \frac{3 \times 10^8}{v_g} \text{ m/s.}$$
$$v_g = \frac{3 \times 10^8}{1.5} \Rightarrow 2 \times 10^8 \text{ ms}^{-1}$$

Thus, the required speed of light in glass = $2 \times 10^8 \text{ ms}^{-1}$.

Q7] 5.0 cm in length is placed at a distance of 20 cm in front of and will be virtual, a convex mirror of radius of curvature 30 cm. Find the position of the image, its nature & size.

Ans 2) Given : $R = 30 \text{ cm}$, $f = R/2 = +15 \text{ cm}$
 $u = -20 \text{ cm}$, $h = 5 \text{ cm}$.

Using the mirror formula:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} - \frac{1}{-20} = \frac{1}{15} + \frac{1}{20} = \frac{4+3}{60}$$

$$\Rightarrow \frac{7}{60}$$

$$v = \frac{60}{7} = 8.57 \text{ cm (approx)}$$

Magnification:

$$m = \frac{-v}{u} = \frac{-60/7}{-20} = \frac{3}{7}$$

$$h' = m \times h = \frac{3}{7} \times 5 = \frac{15}{7} = 2.14 \text{ cm}$$

Q81 Show by ray diagram the image formation by a concave mirror when the object is placed (a) at infinity, (b) beyond the centre of curvature (c) at centre of curvature. State the

position, size and nature of image formed in each case.

Ans $\Rightarrow$ Formation of the image by a concave mirror.

a) When the object is placed at infinity.
The image will be formed.

- i) at focus
- ii) dot size
- iii) inverted
- iv) real.

b) When object is placed beyond the centre of curvature.

The image will be formed:

- i) between focus (F) and centre of curvature (C)
- ii) diminished.
- iii) inverted.
- iv) real.

c) When object is placed at centre of curvature.

The image will be formed:-

- i) at centre of curvature
- ii) size equal to the object.
- iii) inverted.
- iv) real.